

```
data = { 'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40], 'City': ['New York', 'Los Angeles', 'Chicago', 'Houston'] }
```

```
import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, 30, 35, 40],
    'City': ['New York', 'Los Angeles', 'Chicago', 'Houston']
}
df=pd.DataFrame(data)
print(df)
```

	Name	Age	City
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Charlie	35	Chicago
3	David	40	Houston

From the above DataFrame, filter out the rows where the age is greater than 30.

```
filtered_df=df[df['Age']>30]
print(filtered_df)
```

	Name	Age	City
2	Charlie	35	Chicago
3	David	40	Houston

Add a new column Salary with values [50000, 60000, 70000, 80000] to the existing DataFrame.

```
df['Salary']=[50000, 60000, 70000, 80000]
print(df)
```

	Name	Age	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Los Angeles	60000
2	Charlie	35	Chicago	70000
3	David	40	Houston	80000

Increase the salary of all people by 10%.

```
df['Salary']=df['Salary'] * 1.10
print(df)
```

	Name	Age	City	Salary
0	Alice	25	New York	55000.0
1	Bob	30	Los Angeles	66000.0
2	Charlie	35	Chicago	77000.0
3	David	40	Houston	88000.0

Remove the City column from the DataFrame.

```
df1=df.drop('City', axis=1 ,inplace=False)
print(df1)
```

	Name	Age	Salary
0	Alice	25	55000.0
1	Bob	30	66000.0
2	Charlie	35	77000.0
3	David	40	88000.0

Rename the column Age to Years.

```
df.rename(columns={'Age': 'Years'}, inplace=True)
print(df)
```

	Name	Years	City	Salary
0	Alice	25	New York	55000.0
1	Bob	30	Los Angeles	66000.0
2	Charlie	35	Chicago	77000.0
3	David	40	Houston	88000.0

Sort the DataFrame by Salary in descending order.

```
df_sorted = df.sort_values(by='Salary', ascending=False)
print(df_sorted)
```

	Name	Years	City	Salary
3	David	40	Houston	88000.0
2	Charlie	35	Chicago	77000.0
1	Bob	30	Los Angeles	66000.0
0	Alice	25	New York	55000.0

data = { 'Department': ['HR', 'IT', 'HR', 'IT'], 'Employee': ['Alice', 'Bob', 'Charlie', 'David'],
'Salary': [50000, 60000, 55000, 65000] } Group by department and find the average salary.

```
import pandas as pd
data = { 'Department': ['HR', 'IT', 'HR', 'IT'],
        'Employee': ['Alice', 'Bob', 'Charlie', 'David'],
        'Salary': [50000, 60000, 55000, 65000] }
df=pd.DataFrame(data)
grouped=df.groupby('Department')['Salary'].mean()
print(grouped)
```

```

Department
HR      52500.0
IT      62500.0
Name: Salary, dtype: float64

```

```

df1 = pd.DataFrame({ 'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie'] }) df2 =
pd.DataFrame({ 'ID': [1, 2, 4],
'Score': [85, 90, 88] }) Merge them on the ID column.

```

```

df1 = pd.DataFrame({ 'ID': [1, 2, 3],
                      'Name': ['Alice', 'Bob', 'Charlie'] })
df2 = pd.DataFrame({ 'ID': [1, 2, 4],
                      'Score': [85, 90, 88] })
merged=pd.merge(df1,df2,on='ID')
print(merged)

```

	ID	Name	Score
0	1	Alice	85
1	2	Bob	90

Create a DataFrame with some missing values and fill the missing data with appropriate default values.

```

import pandas as pd

data = {
    "Name": ["Alice", "Bob", None, "David"],
    "Age": [21, None, 23, 24],
    "Marks": [85, 90, None, 88]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

df["Name"] = df["Name"].fillna("Unknown")
df["Age"] = df["Age"].fillna(0)
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

print("\nDataFrame after filling missing values:")
print(df)

```

```

Original DataFrame:
   Name  Age  Marks
0  Alice  21.0   85.0
1   Bob   NaN   90.0
2  None  23.0   NaN
3  David  24.0   88.0

```

```

DataFrame after filling missing values:
   Name  Age  Marks
0  Alice  21.0   85.0
1   Bob   0.0   90.0
2  None  23.0   86.5
3  David  24.0   88.0

```

0	Alice	21.0	85.000000
1	Bob	0.0	90.000000
2	Unknown	23.0	87.666667
3	David	24.0	88.000000

Double-click (or enter) to edit